

Real-Time MDM Transmission Trial Using 800G OTN Transceivers over Field-Deployed FMF

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Abstract Through effective suppression of modal crosstalk including adopting bidirectional transmission and only selecting LP_{01} and LP_{02} modes as active channels, we demonstrate real-time MDM-WDM transmission over 80-km field-deployed FMF using commercial 800G DP-16QAM OTN transceivers. ©2026 The Author(s)

Introduction

Mode-division-multiplexing (MDM) technique utilizing linearly polarized (LP) modes in few-mode fibers (FMFs) is widely considered as a promising candidate to address capacity bottleneck of optical communication systems and networks [1, 2]. Especially, the weakly-coupled MDM approaches enable independent signal transmission for each LP mode by significantly suppressing modal crosstalk along the entire FMF link, through which the computation complexity at the receiver could be greatly reduced to at most 4×4 multiple-input multiple-output digital signal processing (MIMO-DSP) [3]. Moreover, the sparse-MDM scheme only utilizing non-degenerate LP modes could employ conventional single-mode transceivers using 2×2 MIMO-DSP [4]. Real-time $LP_{01}/LP_{02}/LP_{03}$ MDM and wavelength-division-multiplexing (WDM) transmission over 300 km FMF using 200G optical transport network (OTN) transceivers without hardware or software modification has been experimentally demonstrated [5]. Recent real-time LP_{01}/LP_{02} MDM transmission based on field-deployed FMF link utilizing commercial single-mode 400G OTN transceivers has been

successfully verified [6].

However, the modal crosstalk in such FMF link should be further suppressed to enhance the optical signal-to-noise (OSNR) to support optical transceivers with higher bit rates beyond 400G or extend the transmission reach [7]. In this paper, we apply bidirectional transmission technique for different LP modes to achieve ultra-low modal crosstalk in the weakly-coupled sparse-MDM transmission scheme [8]. The characteristics of a deployed FMF link for both unidirectional and bidirectional transmission are compared. Then, offline bidirectional $LP_{01}/LP_{21}/LP_{02}$ MDM transmission using 80-GBaud probabilistic constellation shaped (PCS) 16-QAM signals is experimentally demonstrated. Finally, real-time bidirectional LP_{01}/LP_{02} MDM transmission utilizing commercial single-mode single-carrier 800G OTN transceivers with 4 wavelengths is successfully demonstrated. This work verifies for the first time the feasibility of employment for commercial 800G OTN transceivers based on field-deployed FMF.

Characteristics of Deployed Fiber Cable and

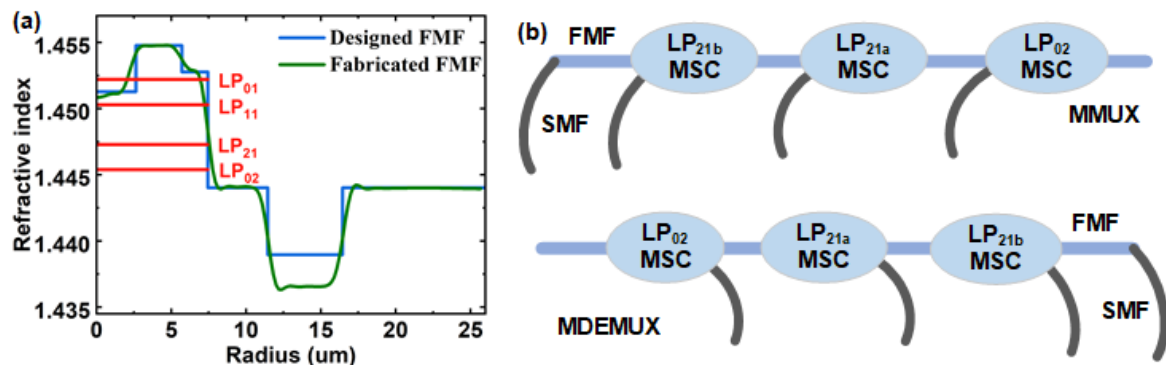


Fig. 1 (a) Index profiles of the 4-LP-mode MRC-FMF. (b) Schematic structures of MMUX and MDEMUX

Mode Multiplexers

Figure 1(a) illustrates the design and fabricated profiles of the 4-LP-mode multi-ring-core (MRC) FMF in the deployed cable [9]. The cabling process strictly adheres to conventional single-mode fiber (SMF) standards. Fig. 1(b) shows the schematic structures of the all-fiber mode multiplexer (MMUX) and mode demultiplexer (MDEMUX) used in this trial, which consists of cascaded mode-selective couplers (MSCs) fabricated by side-polishing and mating processes [10]. The center alignment method is adopted for the conversion of LP_{01} mode. The link is deployed between the Pengcheng Laboratory (PCL) and the Fuyong Laboratory (FYL) in Guangdong province, passing via a transfer node at Binghai Laboratory (BHL). In this experiment, a loopback configuration is employed at the FYL node to achieve a total transmission distance of 80 km over the field-deployed FMF. It should be noted that because of the accumulation of lateral stress in the cabling and layout processing, the LP_{11} mode is excluded from the trial due to its excessive crosstalk with other modes [6].

The characteristics of the FMF link are investigated for two kinds of experimental configurations. There are 35 splicing points in the link. The average propagation losses are measured as 0.212, 0.219, and 0.221 dB/km for the LP_{01} , LP_{21} , and LP_{02} modes, respectively, while the average splicing losses are 0.012, 0.035, and 0.07 dB for the corresponding modes. For the offline trial, forward non-degenerate

LP_{01}/LP_{02} and backward degenerate LP_{21} modes are utilized. The measured bidirectional modal crosstalk reaches -9.42, -15.08, and -11.15 dB for LP_{01} , LP_{21} , and LP_{02} , respectively, whereas the corresponding unidirectional values are -7.44, -12.32, and -9.27 dB. For the real-time trial, the forward LP_{01} mode and backward LP_{02} mode are employed, the bidirectional crosstalk is measured to be -16.14 dB for LP_{01} and -17.66 dB for LP_{02} , respectively, while they are -9.80 dB and -14.80 dB, respectively, for the unidirectional case. We can see that the bidirectional transmission scheme could effectively reduce the modal crosstalk and increase the OSNR margin required by higher-order modulation signals.

Experimental setup and results for offline MDM transmission

Figure 2(a) depicts the experimental setup for bidirectional offline MDM transmission over the deployed fiber link. At the transmitter, an optical carrier, generated by an external cavity laser (ECL), is modulated by a dual-polarization IQ modulator (DP-IQ-MOD) to produce an 80-GBaud Nyquist PCS-16QAM signal with 3 bits/symbol entropy. The signal is then combined with three ASE-based dummy channels via a wavelength-selective switch (WSS). The optical spectra of the four 80-GBaud WDM channels is displayed in Fig. 2(b). The combined WDM signal is amplified by an erbium-doped fiber amplifier (EDFA) and subsequently split into four paths by a 1×4 coupler, with relative fiber delays applied

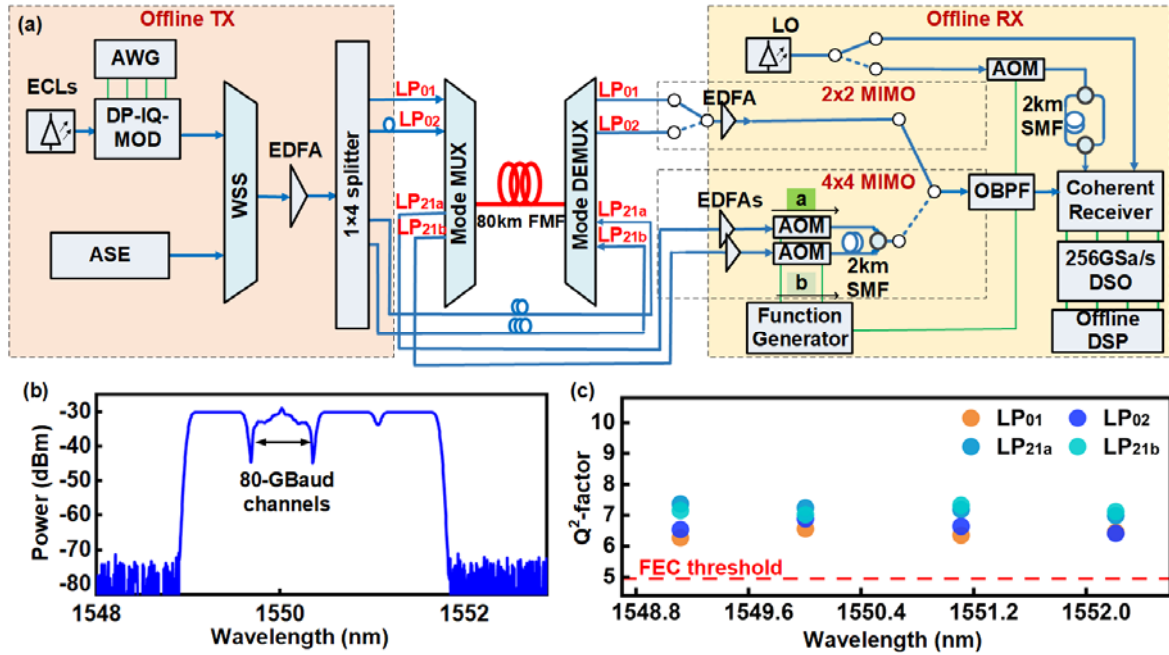


Fig. 2 (a) Experimental setup for offline bidirectional MDM transmission. (b) Optical spectrum at 1550 nm. (c) Transmission performance of 4 wavelengths. ECL: external-cavity laser. DP-IQ-MOD: dual-polarization in-phase and quadrature modulator. AWG: arbitrary waveform generator (Keysight M8199A). ASE: amplified spontaneous emission. WSS: wavelength-selective switch. EDFA: erbium-doped fiber amplifier. AOM: acoustic-optic modulators. OBPF: optical band-pass filter. DSO: digital storage oscilloscope (Keysight UXR0594AP).

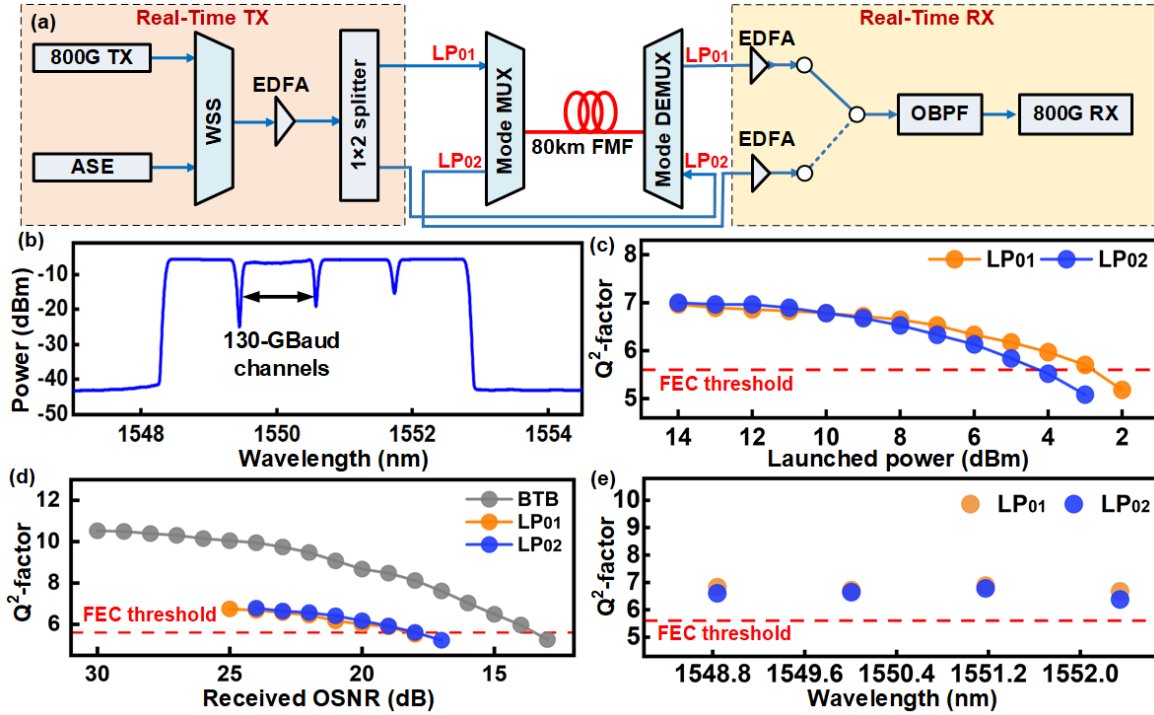


Fig. 3 (a) Experimental setup for real-time MDM transmission. (b) Optical spectrum at 1550 nm. (c) Transmission performance versus launched power. (d) Transmission performance versus different OSNR. (e) Transmission performance of 4 wavelengths

for decorrelation. To implement the bidirectional link, two signal paths are launched into the LP₀₁ and LP₀₂ ports of the MMUX for forward transmission, while the remaining two are injected into the LP_{21a} and LP_{21b} ports of the MDEMUX for reverse transmission. At the receiver, a time-domain multiplexed (TDM) reception strategy is utilized, so that single 4-channel digital storage oscilloscope (DSO) could capture the waveforms from both the degenerate LP₂₁ modes [11]. During offline DSP, 2×2 MIMO-DSP is applied to the non-degenerate LP₀₁ and LP₀₂ modes, whereas 4×4 MIMO-DSP is applied to the LP₂₁ mode. The measured performance is shown in Fig. 2(c). We can see that the Q²-factor for all wavelength channels in all the LP modes are above 4.95 dB, which is the threshold with 25.5% soft-decision forward error correction (SD-FEC) overhead [12]. The minimum Q²-factor of 6.27 dB is observed for the LP₀₁ mode at 1549.11 nm.

Experimental setup and results for real-time MDM transmission

Figure 3(a) illustrates the experimental setup for real-time bidirectional MDM transmission utilizing commercial 800G OTN transceivers without hardware or software modifications, which operates with the modulation format of 130-GBaud 16-QAM. At the transmitter, the test signal from the 800G transmitters is combined with three ASE dummy channels via a WSS. The optical spectra of the four WDM channels are

captured after the WSS and shown in Fig. 3(b). The combined WDM signal is amplified by an EDFA and subsequently split into two paths by a 1×2 coupler. To implement the bidirectional link, the signals are launched into the LP₀₁ port of the MMUX for forward transmission and the LP₀₂ port of the MDEMUX for reverse transmission, respectively. At the receiver, the signals are amplified by EDFAs and filtered by optical band-pass filters (OBPFs). Fig. 3(c) and (d) show the Q²-factor performance versus the launched optical power and the received OSNR. The transmission results for the four WDM channels across both modes are presented in Fig. 3(e). The Q²-factor for all 2-mode 4-wavelength channels are above the 5.6 dB FEC threshold of the OTN transceivers. The worst performance of 6.38 dB is observed at 1552.35 nm in the LP₀₂ mode.

Conclusion

We demonstrate real-time LP₀₁/LP₀₂ bidirectional MDM transmission over 80-km field-deployed weakly-coupled FMF. By effective suppression of modal crosstalk, commercial 800G OTN transceivers are employed without hardware or software modification.

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References

- [1] D. J. Richardson, J. M. Fini, and L. E. Nelson, "Space-division multiplexing in optical fibres," *Nature Photonics*, vol. 7, no. 5, pp. 354–362, 2013. DOI: [10.1038/nphoton.2013.94](https://doi.org/10.1038/nphoton.2013.94).
- [2] G. Li, N. Bai, N. Zhao, and C. Xia, "Space-division multiplexing: the next frontier in optical communication," *Advances in Optics & Photonics*, vol. 6, no. 4, pp. 413–487, 2014. DOI: [10.1364/AOP.6.000413](https://doi.org/10.1364/AOP.6.000413).
- [3] S. Beppu, D. Soma, S. Sumita, Y. Wakayama, H. Takahashi, T. Tsuritani, I. Morita, and M. Suzuki, "402.7-Tb/s MDM-WDM Transmission Over Weakly Coupled 10-Mode Fiber Using Rate-Adaptive PS-16QAM Signals," *Journal of Lightwave Technology*, vol. 38, no. 10, pp. 2744–2750, 2020. DOI: [10.1109/JLT.2020.2979195](https://doi.org/10.1109/JLT.2020.2979195).
- [4] G. Qiao, Y. Yang, H. Ji, S. Huang, C. Long, Y. Gao, M. Zuo, J. Zhang, Z. Xu, Q. Wu, S. Wang, L. Liu, L. Shen, J. Luo, Z. He, Y. He, W. Hu, Z. Chen, and J. Li, "205.8-Tb/s Weakly-Coupled 2-Mode 7-Core Transmission Over 1170-km FM-MCF Only Using 2×2 MIMO-DSP," *Journal of Lightwave Technology*, vol. 43, no. 4, pp. 1934–1940, 2025. DOI: [10.1109/JLT.2024.3496519](https://doi.org/10.1109/JLT.2024.3496519).
- [5] M. Zuo, D. Ge, Y. Gao, J. Cui, S. Huang, R. Zhou, Q. Guo, Y. Zhang, D. Zhang, X. Xiao, L. Shen, D. Wang, Y. Li, L. Han, L. Zhang, X. Lan, D. Zhang, H. Li, Y. He, Z. Chen, and J. Li, "3-mode Real-time MDM Transmission Using Single-mode OTN Transceivers over 300 km Weakly-coupled FMF," in *2022 Optical Fiber Communication Conference (OFC)*, paper M4B.4. DOI: [10.1364/OFC.2022.M4B.4](https://doi.org/10.1364/OFC.2022.M4B.4).
- [6] G. Qiao, L. Feng, A. Zhang, C. Long, L. Liu, H. Ji, T. Ji, Y. Yang, S. Wang, Z. Xu, Y. Gao, J. Yu, L. Shen, L. Zhang, J. Luo, X. Zhang, Z. He, Y. He, Z. Chen, H. Li, W. Hu, S. Yu, and J. Li, "First Real-Time MDM Transmission Trial over Field-Deployed Weakly-Coupled Few-Mode Fiber," in *Asia Communications and Photonics Conference (ACP)*, pp. 1–4, 2025. DOI: [10.1109/ACP66871.2025.11350949](https://doi.org/10.1109/ACP66871.2025.11350949).
- [7] G. Qiao, M. Zuo, J. Zhang, C. Long, Y. He, Z. Chen, and J. Li, "Statistical Characteristics of Distributed Modal Crosstalk for Weakly-Coupled FMF Transmission," in *Conference on Lasers and Electro-Optics (CLEO)*, Optica Publishing Group, 2023, paper SF1M.4. DOI: [10.1364/CLEO_SI.2023.SF1M.4](https://doi.org/10.1364/CLEO_SI.2023.SF1M.4).
- [8] J. Zhang, X. Wu, Q. Fan, X. Yi, Z. Tan, J. Li, Z. Li, and C. Lu, "High-capacity bi-directional full-duplex transmission based on fiber-eigenmode multiplexing over a FMF with 2×2 MIMO," *Optics Express*, vol. 29, no. 19, pp. 30473–30482, 2021. DOI: [10.1364/OE.433972](https://doi.org/10.1364/OE.433972).
- [9] S. Huang, L. Shen, G. Qiao, Y. Ding, Y. Gao, J. Cui, B. Zhu, S. Liu, M. Zuo, J. Zhu, L. Zhang, J. Luo, Y. He, Z. Chen, and J. Li, "Low-modal-crosstalk doped-fiber amplifiers in few-mode-fiber-based systems," *Photonics Research*, vol. 12, no. 8, pp. 1768–1775, 2024. DOI: [10.1364/PRJ.521376](https://doi.org/10.1364/PRJ.521376).
- [10] C. Long, J. Cui, Y. Gao, G. Qiao, B. Zhu, J. Zhang, Y. Yang, L. Shen, J. Luo, Y. He, Z. Chen, and J. Li, "Ultra-Wideband Mode Selective Couplers for Weakly-Coupled WDM-MDM Transmission," in *2024 Optical Fiber Communications Conference and Exhibition (OFC)*, 2024, paper M2E.5. DOI: [10.1364/OFC.2024.M2E.5](https://doi.org/10.1364/OFC.2024.M2E.5).
- [11] M. Zuo, D. Ge, J. Liu, Y. Gao, L. Shen, X. Lan, Z. Chen, Y. He, and J. Li, "Long-haul intermodal-MIMO-free MDM transmission based on a weakly coupled multiple-ring-core few-mode fiber," *Optics Express*, vol. 30, no. 4, pp. 5868–5878, 2022. DOI: [10.1364/OE.451971](https://doi.org/10.1364/OE.451971).
- [12] K. Sugihara, Y. Miyata, T. Sugihara, K. Kubo, H. Yoshida, W. Matsumoto, and T. Mizuochoi, "A Spatially-coupled Type LDPC Code with an NCG of 12 dB for Optical Transmission beyond 100 Gb/s," in *Optical Fiber Communication Conference/National Fiber Optic Engineers Conference (OFC/NFOEC)*, 2013, paper OM2B.4. DOI: [10.1364/OFC.2013.OM2B.4](https://doi.org/10.1364/OFC.2013.OM2B.4).